

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester VI Examination, 2022-23

Zoology

Paper: ZOB-604 — Evolution and Animal Behaviour

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A (including Question No. 1 which is compulsory) and three questions from Section-B (including Question No. 1 which is compulsory). Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A
(Evolution) (Marks: 35)

Question 1. Answer the following parts:

(a) Define the following:

[5 × 1 = 5]

(i) Assortative mating

(ii) SNP

(iii) Allozyme

(iv) Eon

(v) Sympatric speciation.

(b) Differentiate between the following:

[3 × 2 = 6]

(i) Relative dating and Radioactive dating

(ii) Founder effect and Bottleneck effect

(iii) Concealing and Alluring mimicry.

Question 2. Write short notes on the following:

[3 × 4 = 12]

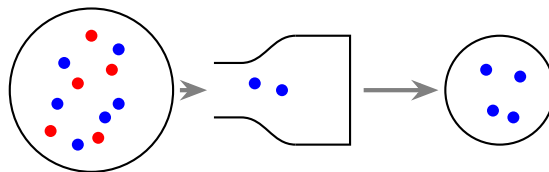
(a) Tandem repeat polymorphism

(b) The chronological evolution of horse

(c) Natural selection as mode of evolution.

Question 3. Giving suitable examples, explain various isolating mechanisms that hinder interbreeding between the species. [12]

Question 4. Give a brief account of different types of fossils. Add a note on the significance of fossil study. [12]



SECTION – B
(Animal Behaviour) (Marks: 35)

Question 1. Answer the following parts:

(a) State whether the following statements are True or False. Justify your answer in 2-3 sentences each: [3 × 2 = 6]

- (i) A fixed action pattern is a changeable series of actions triggered without any stimulus.
- (ii) Learned behaviour is inherited, automatic and inflexible.
- (iii) An animal making new associations between previously learned tasks to solve a new problem is known as latent learning.

(b) Define the following: [5 × 1 = 5]

- (i) Habituation
- (ii) Imprinting
- (iii) Inclusive fitness
- (iv) Circadian rhythm
- (v) Zugunruhe.

Question 2. Describe the patterns of different social behaviours in animals. [12]

Question 3. Answer the following:

- (a) Write a note on genetic control of behaviour. [6]
- (b) Describe different modes of communication in animals. [6]

Question 4. Write notes on the following: [2 × 6 = 12]

- (a) Costs and benefits of group living
- (b) Migration in birds.

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